TECHNICAL SPECIFICATION & AI ARCHITECTURE BLUEPRINT

Intelligence Sales CRM

A comprehensive technical and functional handbook for Data Scientists, AI Engineers, and Fullstack Developers. Explains the commercial system from Executive, Sales Rep, and B2B Client perspectives, accompanied by live platform screenshots and rigorous architectural specifications for all 7 AI Agents.

TARGET AUDIENCE	SYSTEM VERSION	ARCHITECTURE	AI AGENTS COUNT
Data / AI Dev Team	v3.2 Enterprise (2026)	React + Laravel + AI Hub	7 Autonomous AIs

Primary Goal of this Blueprint

This document serves as the binding contract for the Data/AI developer to build, train, prompt, and deploy the 7 commercial AI agents. It details exact trigger hooks, input JSON schemas, prompt logic, output database mutations, and visual links to every corresponding interface in the Intelligence Sales platform.

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Generated: October 2026 • Production Ready

1. System Architecture & Core Philosophy

Core Principle: Intelligence Sales is *not* an AI novelty chatbot. It is a high-velocity B2B Commercial Operating System engineered to drive real wholesale and enterprise product trade. The 7 AI Agents function as autonomous force multipliers embedded inside human sales workflows—handling lead scoring, margin protection, follow-up automation, and contract generation while keeping humans in the loop for final commitment.

1.1 Core Domain Entities & Relational Hierarchy

The database layer consists of 7 tightly interconnected relational entities mapped inside Laravel 11 with SQLite/MySQL:

ENTITY	DATABASE TABLE	KEY ATTRIBUTES	ROLE IN WORKFLOW
Account	`accounts`	name, industry, city, phone, email, status, tier, ICE, IF, Patente, credit_limit	Enterprise client company profile. The parent root for all deals, quotes, and contacts.
User	`users`	name, email, role, account_id, title, phone, quota	System actor. Roles: `admin`, `manager`, `rep`, or `client`.
Lead	`leads`	name, company, email, phone, source, status, ai_score, score_reason, user_id, account_id	Inbound prospect. Evaluated from 0-100 by the Lead Intelligence Agent before conversion.
Deal	`deals`	title, account_id, user_id, amount, stage, probability, expected_close_date	Active revenue opportunity in the 6-stage sales pipeline. Owned by a sales rep.
Quote	`quotes`	quote_number, deal_id, account_id, discount_percent, discount_approved, public_token	Commercial offer with itemized line items, discount governance, and legal E-signature token.
Activity	`activities`	user_id, deal_id, lead_id, account_id, type, title, description, completed, idempotency_key	Unified audit and follow-up timeline tracking every call, email, meeting, and stage shift.

1.2 Role-Based Access Control (RBAC) & Data Isolation

Data scoping is strictly enforced at the database query level via `App\Support\CrmAccess`:

- **Admin & Sales Director:** Full visibility across all reps, all deals, company-wide revenue forecasts, user management, and system configuration.
- **Sales Manager:** Can reassign deals among commercial representatives, supervise team quotas, and holds exclusive authority to approve quotes with discounts exceeding 15%.
- **Commercial Sales Representative (Rep):** Restricted strictly to deals, quotes, and accounts in their assigned portfolio, plus unassigned inbound leads. Reps cannot reassign deals to other users.
- **B2B Client:** Restricted strictly to their own enterprise account records (`account_id`). Can only view proposals addressed to their enterprise and sign quotes.

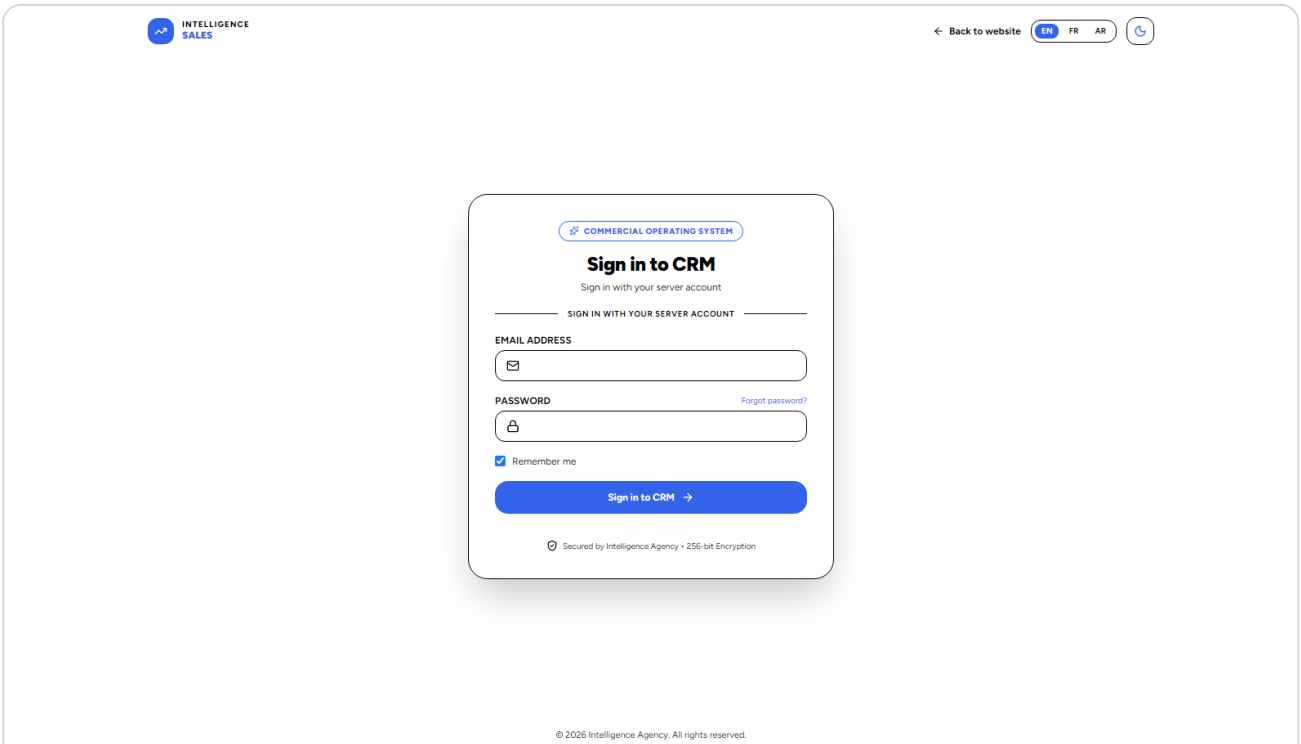


Figure 1: Authentication Portal with multi-role RBAC access switcher (Admin, Manager, Rep, Client). `URL: /login`

2. Perspective 1: Executive & Sales Management (Admin / Manager)

For the CRO, VP of Sales, and Regional Sales Managers, Intelligence Sales provides a centralized mission-control cockpit designed to govern revenue, eliminate deal slippage, inspect rep behaviors, and enforce commercial margin rules.

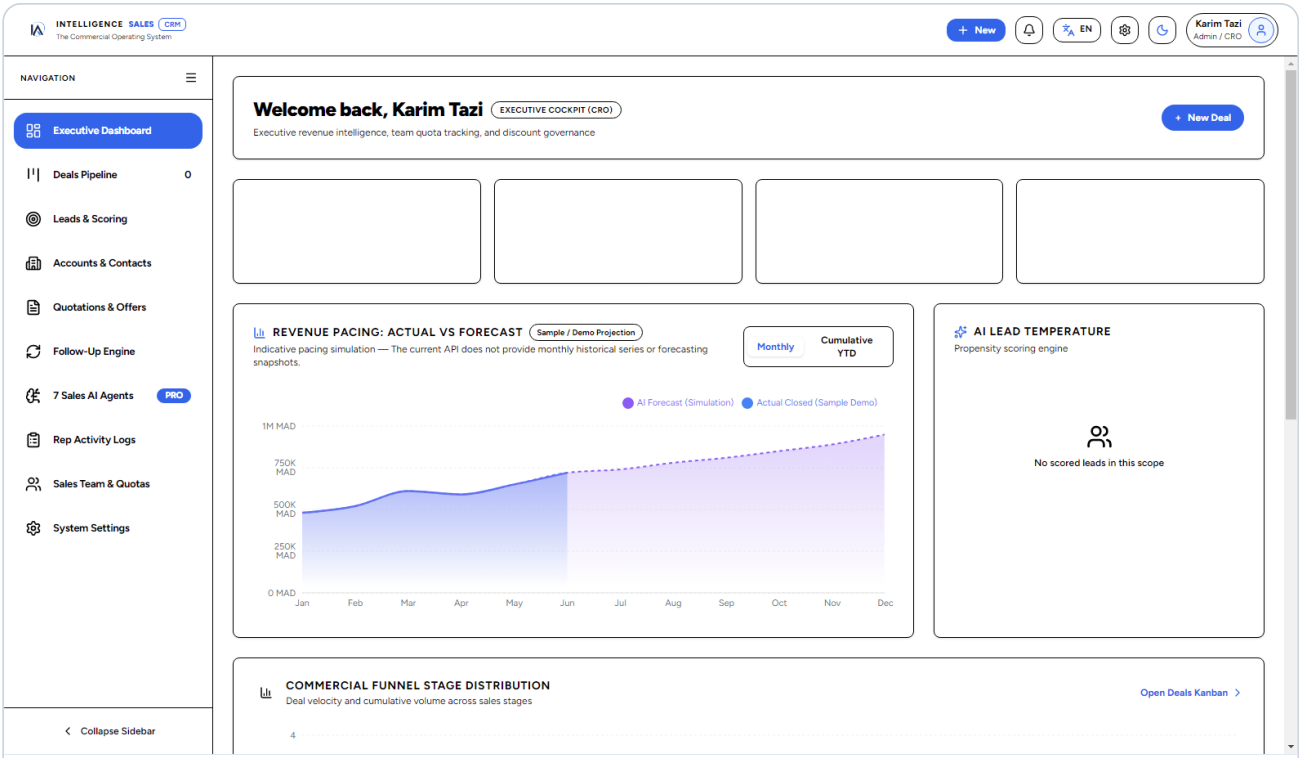


Figure 2: Executive Dashboard — Real-time revenue forecasting, stage conversion funnel, and team quota pacing. `URL: /crm/dashboard`

2.1 Revenue Forecasting & Quota Pacing

The dashboard synthesizes closed-won ARR/MRR against annual commercial targets. Revenue projections are weighted automatically by stage probabilities (Leads 10%, Qualification 30%, Discovery 50%, Proposal 70%, Negotiation 80%, Won 100%).

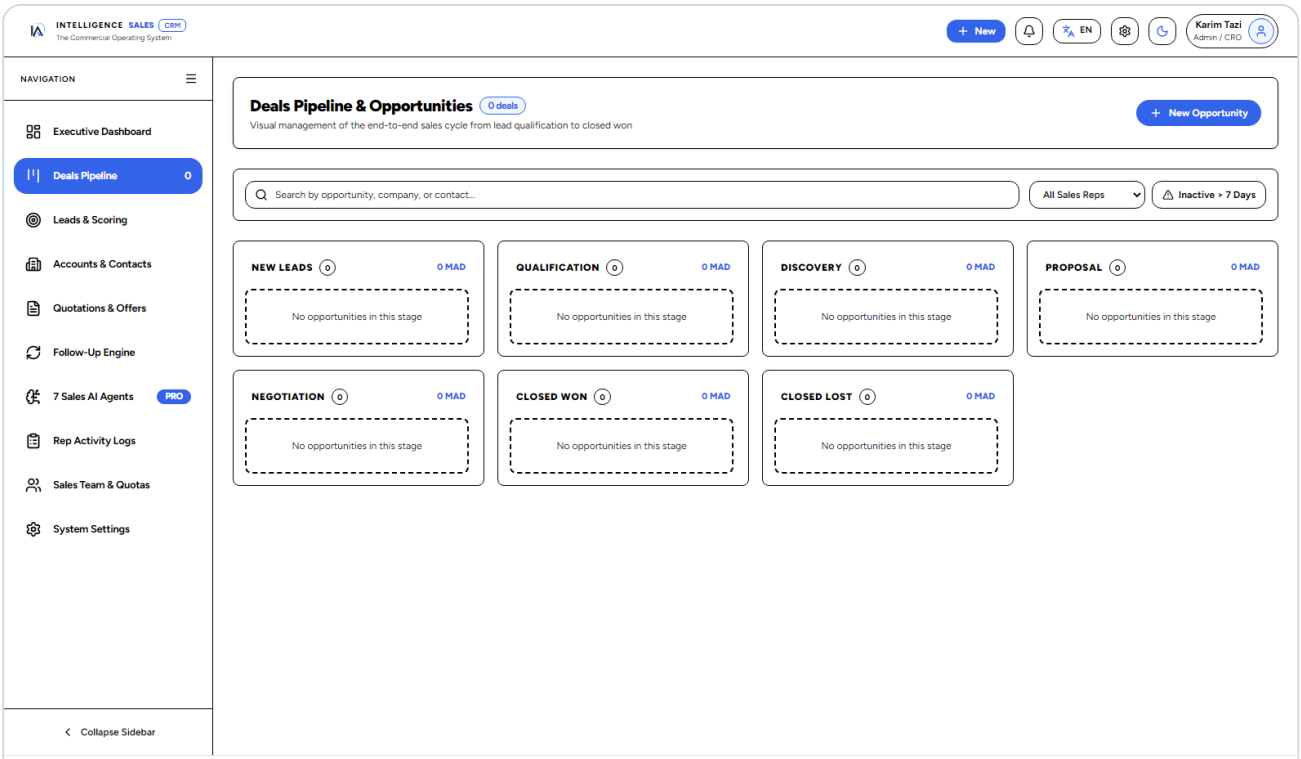


Figure 3: Deals Pipeline Kanban — Multi-stage visual progression with deal values, probabilities, and stage locks. `URL: /crm/pipeline`

2.2 Deal Assignment & Rep Workload Reallocation

Managers can inspect any opportunity card and reassign it to another sales representative via the Deal Inspection Modal. The backend checks caller role via `CrmAccess::isManager()`, preventing unauthorized reps from altering deal ownership while empowering managers to balance team workloads.

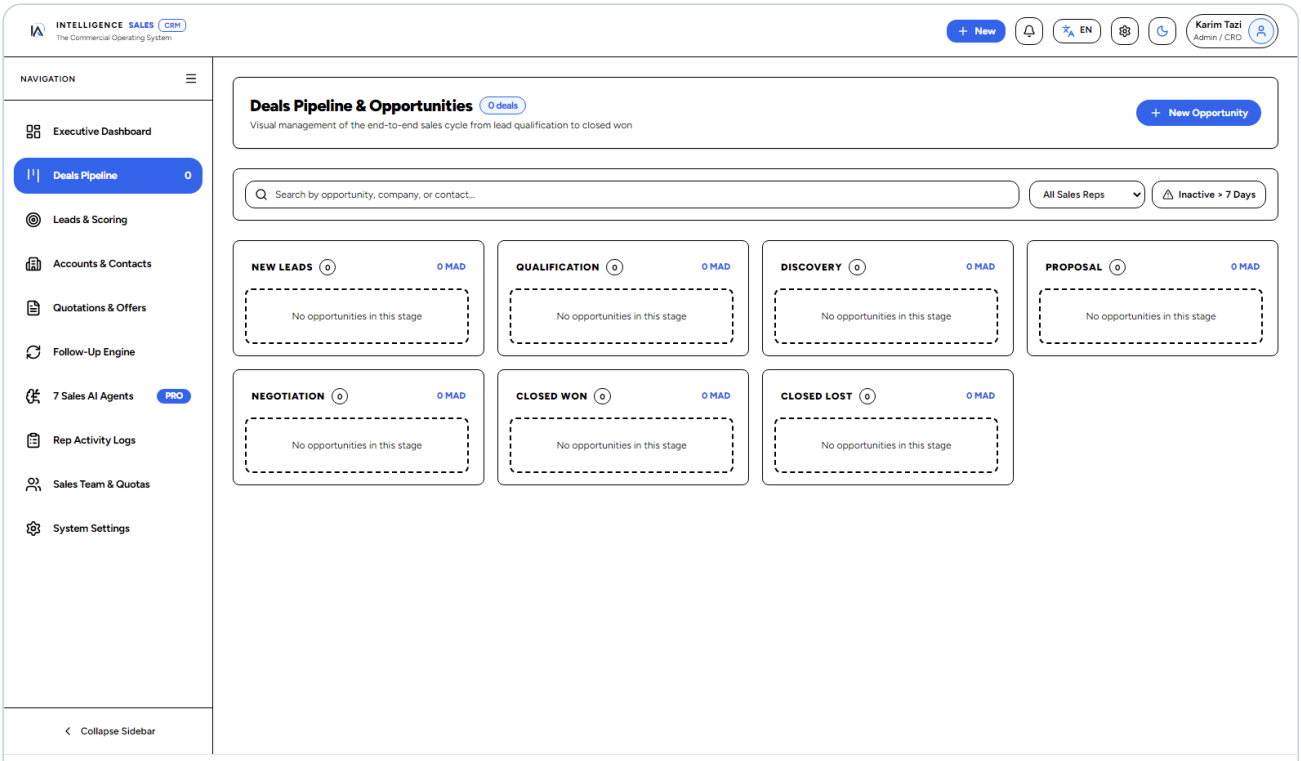


Figure 4: Deal Detail Modal — AI next-step recommendations, product bundles, and Manager-only Rep Selector. `Component: CrmPipeline.jsx`

2. Perspective 1 (Cont.): Governance, Auditability & Margin Protection

2.3 Rep Activity Logs & Audit Trail (/crm/rep-logs)

A dedicated audit navigation item (`/crm/rep-logs`) gives managers complete step-by-step transparency into every action performed by sales representatives: stage advancements, inbound lead conversions, quotation dispatches, and client communications. Each log records the exact rep ID, timestamp, target account, and operational payload.

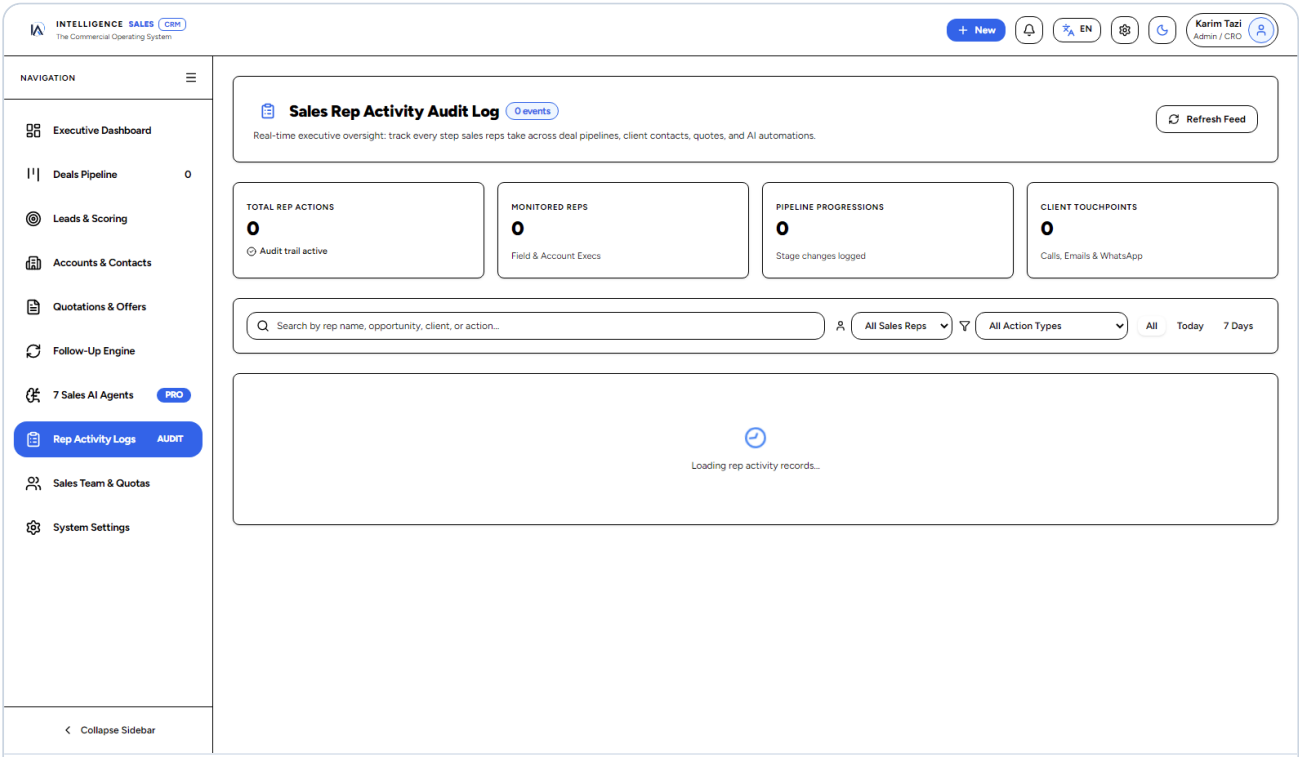


Figure 5: Rep Activity Logs — Audit trail showing real-time timestamps, deal progression, and rep actions. `URL: /crm/rep-logs`

2.4 Margin Protection & The 15% Discount Approval Rule

Sales reps frequently encounter price pushback. However, unconstrained discounting degrades enterprise profit margins. The platform enforces an ironclad governance threshold:

- **Discounts ≤ 15%:** Commercial reps can approve and dispatch directly to the client.
- **Discounts > 15%:** The quotation status locks automatically into **PENDING APPROVAL** . The dispatch button and public signing token are disabled until a Sales Manager or Director clicks `Approve Discount` . Any direct attempt to bypass this via API yields a `403 Forbidden` error.

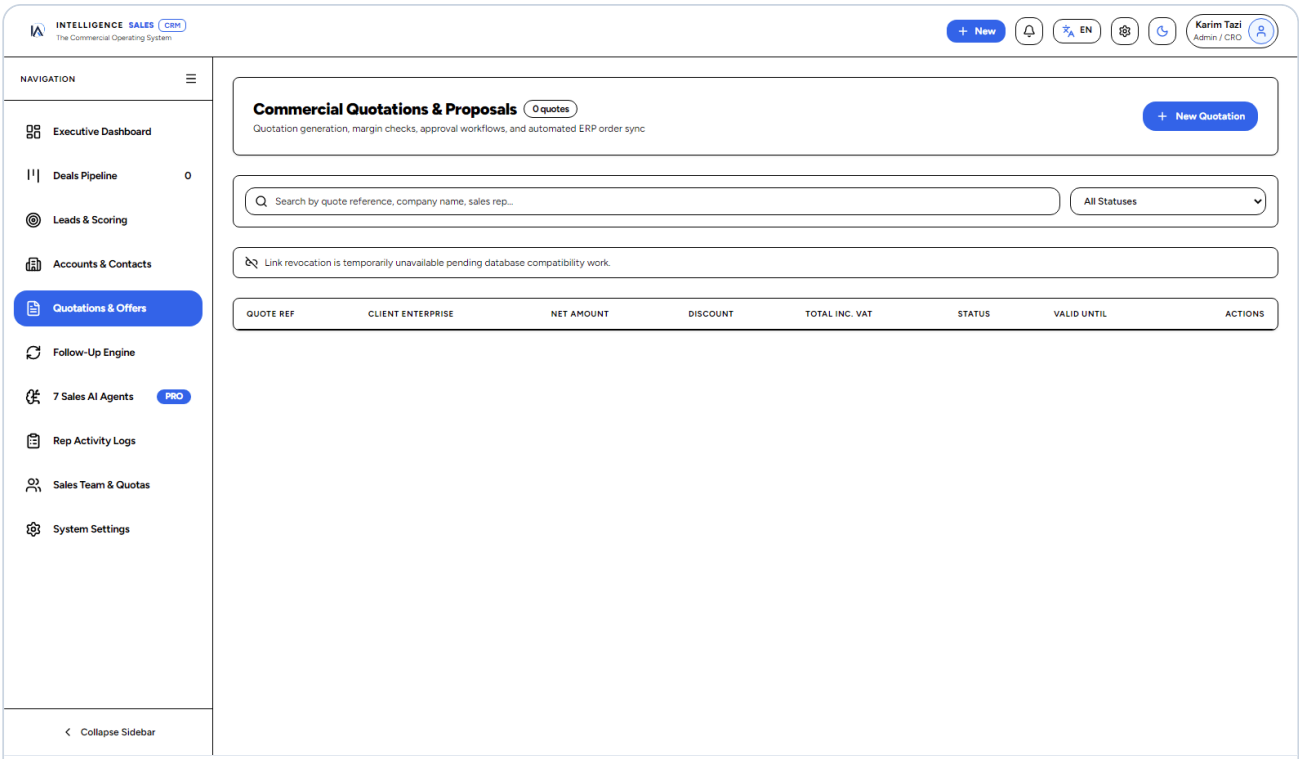


Figure 6: Quotations Directory — Enforcing margin rules, discount approvals, and secret public token links. `URL: /crm/quotes`

3. Perspective 2: Commercial Sales Representative (Rep) Workstation

For the sales executive in the field, Intelligence Sales acts as an intelligent co-pilot. Reps start their day by reviewing assigned inbound leads, checking AI qualification scores, advancing pipeline opportunities, and drafting itemized product quotes.

3.1 Inbound Lead Claiming & AI Qualification Scoring (/crm/leads)

When an inbound proposal request arrives from the website, it enters the rep queue marked with a pulsating **NEW** badge on the card corner. The Lead Intelligence Agent scores the prospect from 0 to 100 based on estimated order volume, corporate domain credibility, and stated requirements.

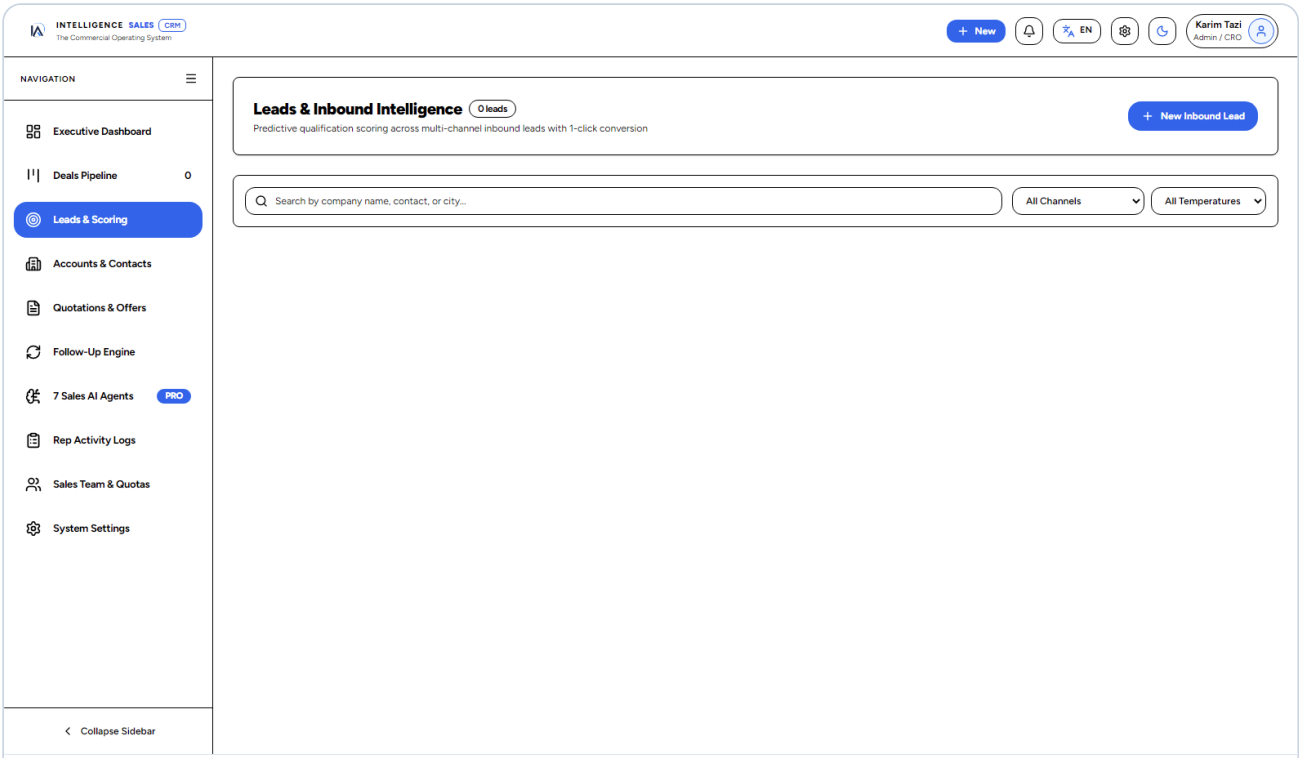


Figure 7: Leads Directory — Pulsating Red "NEW" corner badge, 0-100 AI Score, and 1-Click Convert button. `URL: /crm/leads`

3.2 1-Click Conversion: Lead → Enterprise Account + Opportunity

Clicking `Convert to Opportunity` executes an atomic transaction in MySQL: creates an active Enterprise Account, provisions a Deal in the Qualification stage, links the contact, and archives the lead.

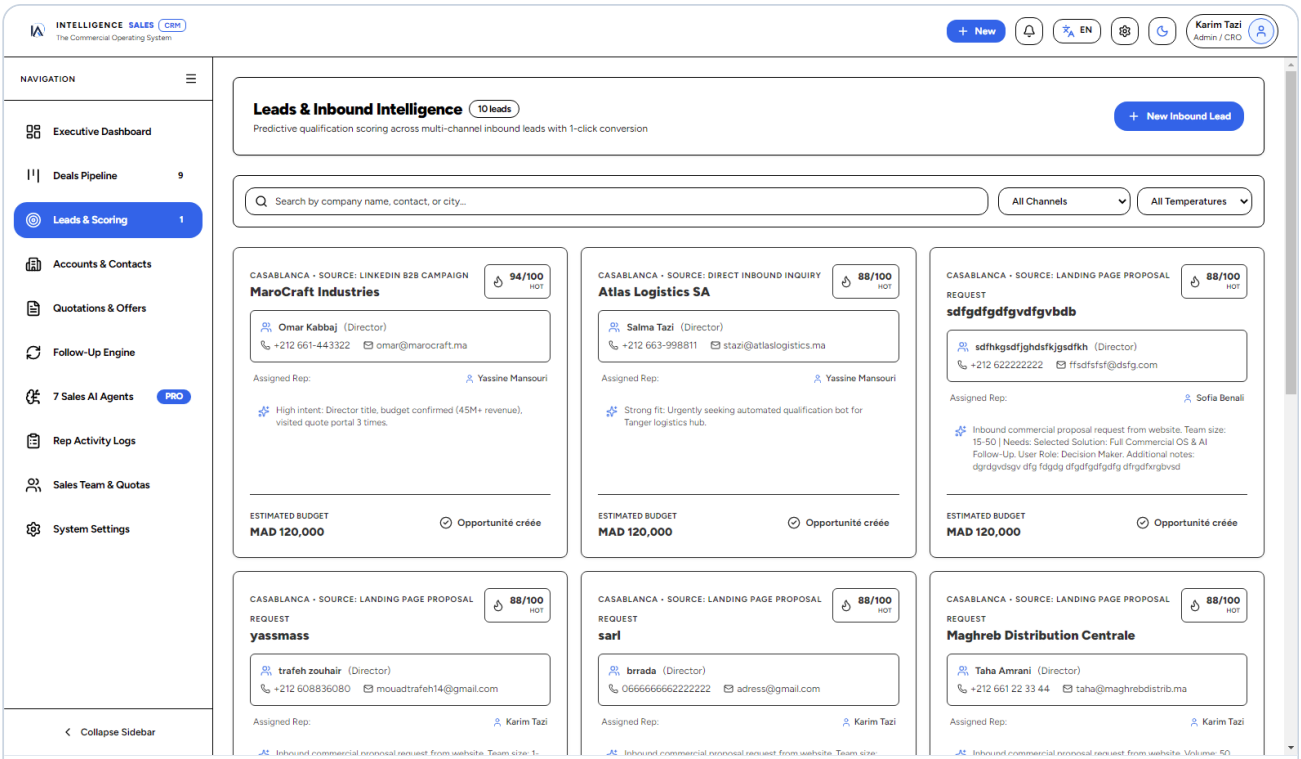


Figure 8: New Lead Modal — Capturing contact info with searchable international dial code and country flag. `Component: PhoneInputField.jsx`

3.3 Smart Follow-Up & The 7-Day Inactivity Rule (/crm/activities)

Deals with no interaction for 7 days are automatically flagged. The Follow-Up Agent drafts personalized re-engagement messages in French, Arabic, and English for one-click rep review before transmission.

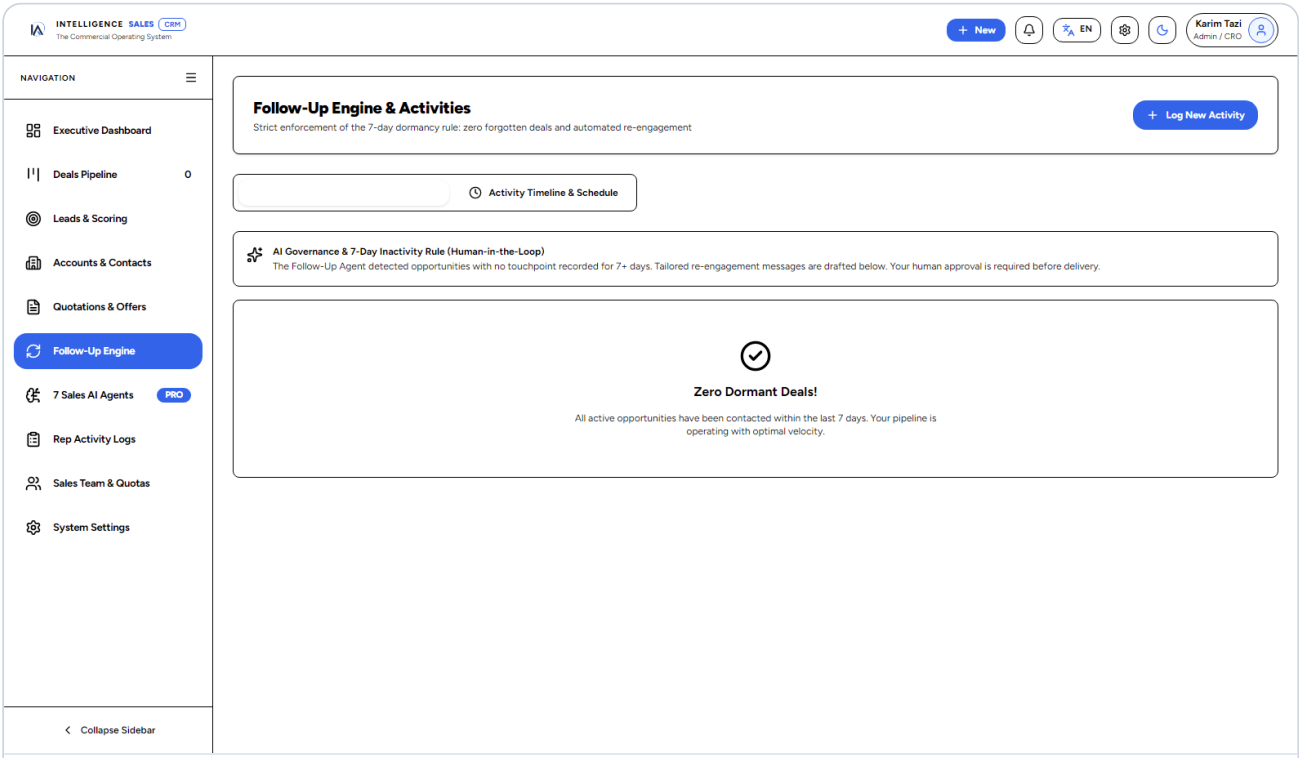


Figure 9: Activities Timeline — 7-day dormancy warnings, scheduled calls, and automated follow-up drafts. `URL: /crm/activities`

4. Perspective 3: B2B Client Experience & Journey

The B2B client experience is frictionless, transparent, and professional—from initial proposal submission on the landing page to portal access and legal digital contract signing.

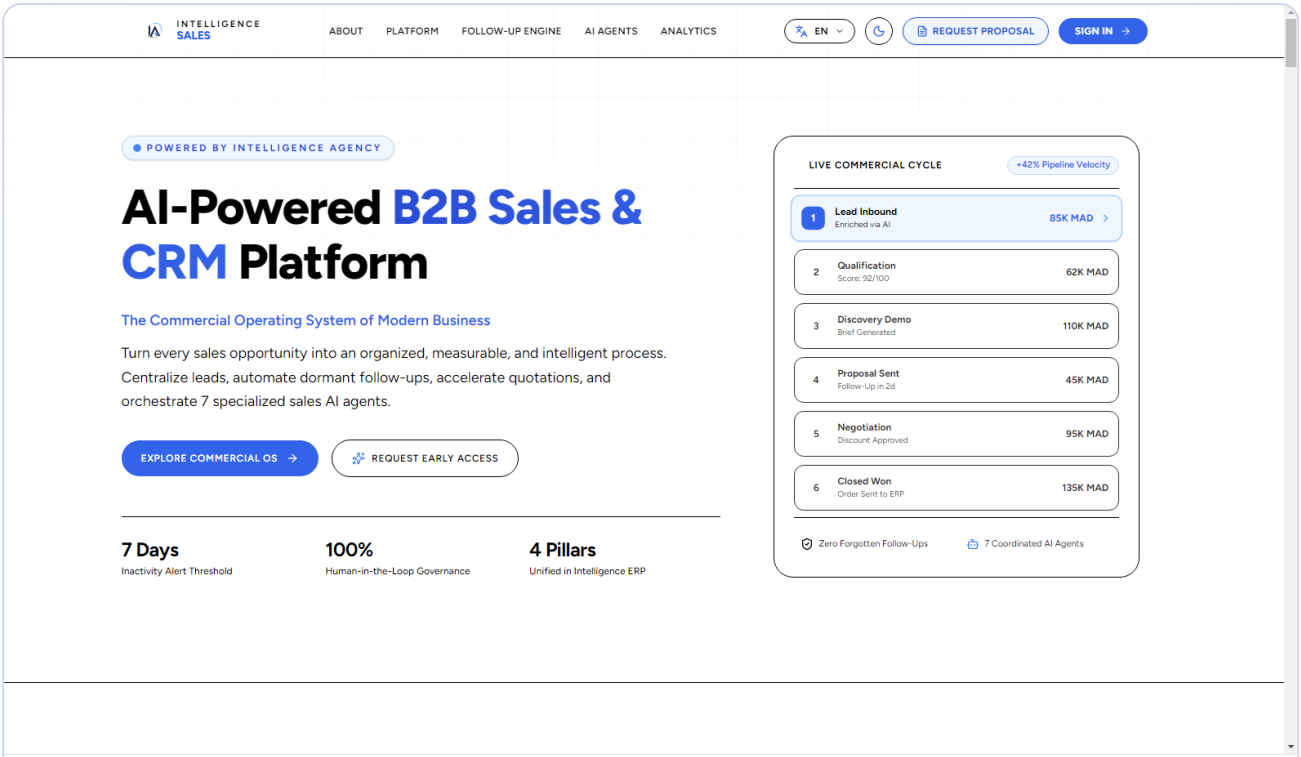


Figure 10: Landing Page Hero — Highlighting wholesale product distribution and Request Proposal CTA. `URL: /`

4.1 Seamless Inbound Quotation Request & Automated Account Creation

When a prospect clicks `Demande de Devis`, they enter their enterprise details and phone number using the international flag selector. Upon submission:

1. An Enterprise **Account** is automatically created in the database.
2. A Client **User** portal account is created with a generated temporary password (e.g. `Client@9750`).
3. A branded Welcome Email is dispatched to the submitted inbox with quotation reference (`DEV-2026-XXXX`), credentials, and portal link.
4. The credentials card appears immediately on screen with a 1-click **Copy Credentials** button and direct **Go to Portal** link.

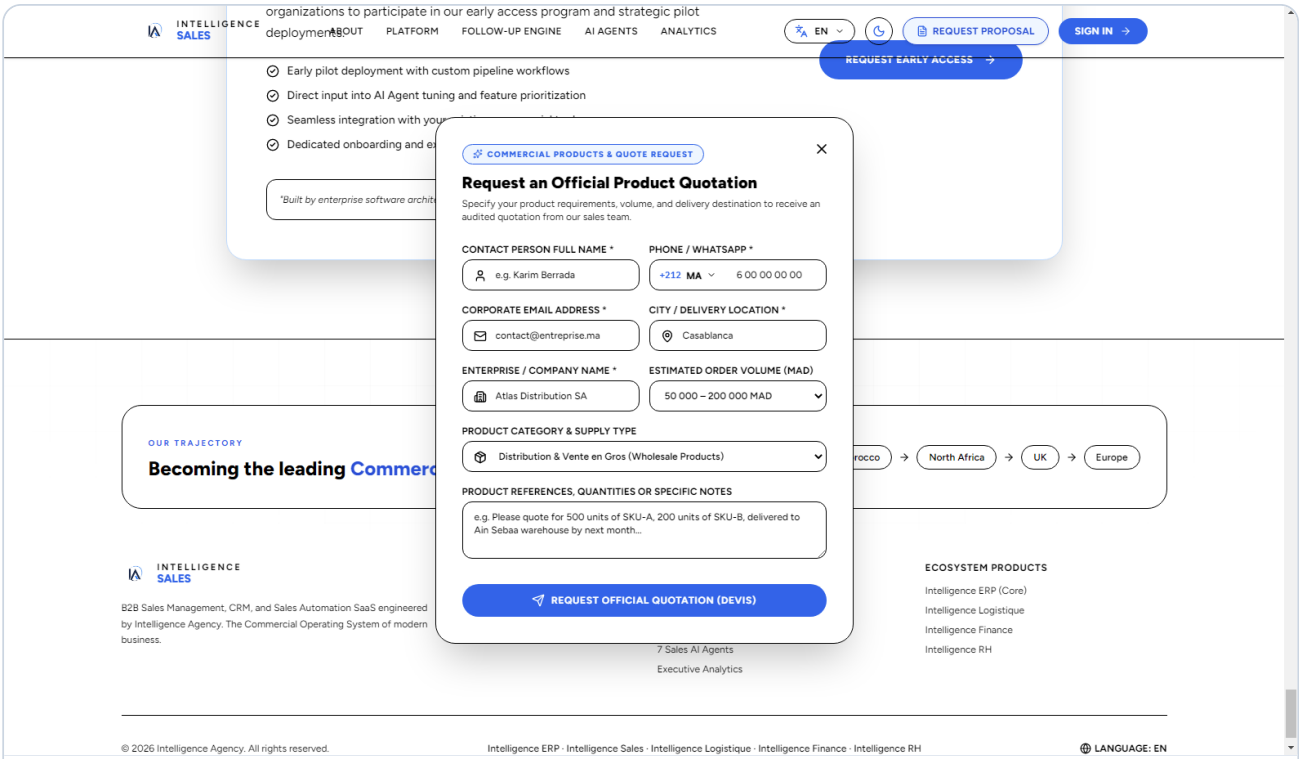


Figure 11: Quotation Request Modal — Phone input starting with +dial code and national flag emoji. `Component: PartnerModal.jsx`

4. Perspective 3 (Cont.): Client Space & Legal E-Signature Portal

4.2 The Dedicated B2B Client Space (/crm/dashboard as Client)

When clients log in, their navigation is strictly scoped to client-facing features: *Mon Espace Client*, *Mes Devis & Offres*, and *Historique & Jalons*. They can inspect proposals addressed to their enterprise and contact their assigned commercial executive directly.

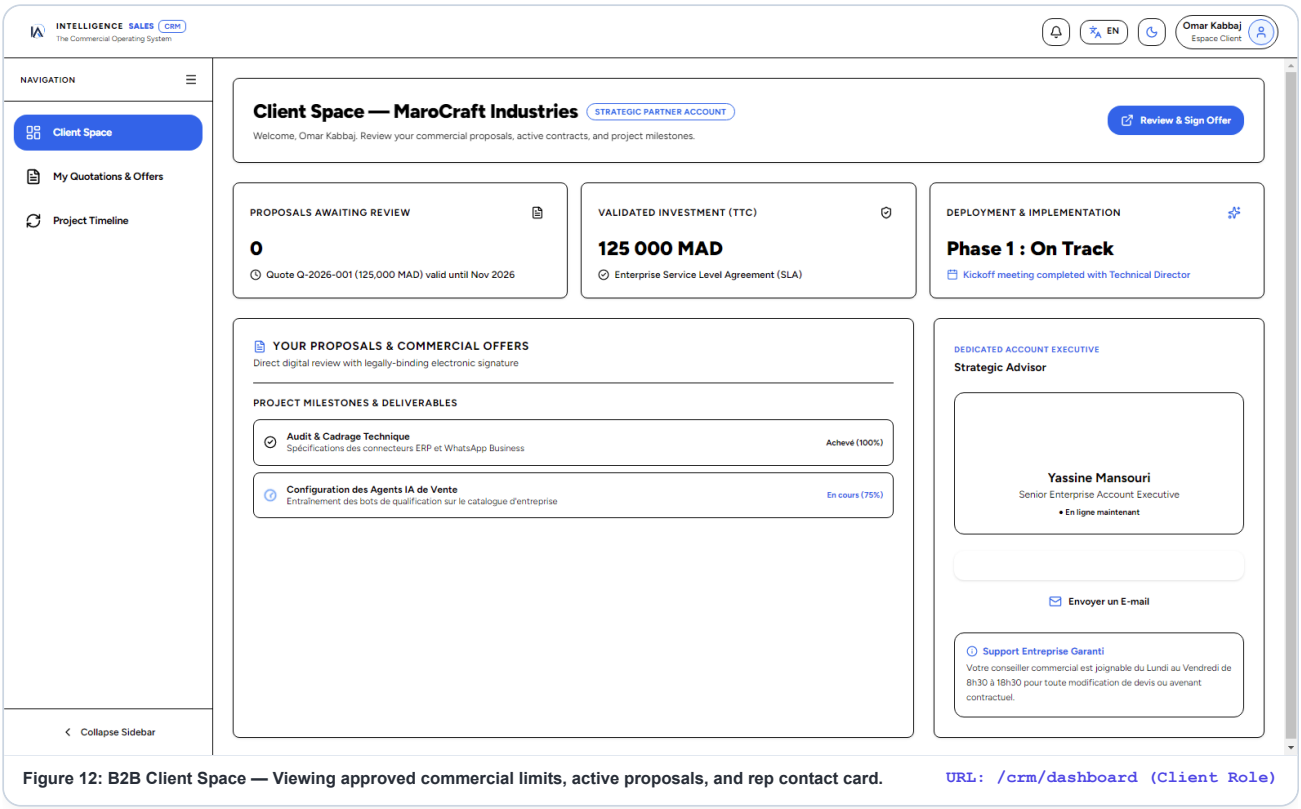


Figure 12: B2B Client Space — Viewing approved commercial limits, active proposals, and rep contact card. `URL: /crm/dashboard (Client Role)`

4.3 Public Quotation Review & Legal E-Signature (/portal/quote/:token)

Clients access their itemized quotation through a secure secret URL. The portal displays full commercial transparency: item descriptions, unit prices, volume discounts, and 20% legal Moroccan VAT (TVA).

Clients can sign electronically by providing their Signer Name, optional Purchase Order (PO) number, special instructions, and agreeing to the formal commercial terms. Signing automatically moves the CRM Deal to **CLOSED WON** .

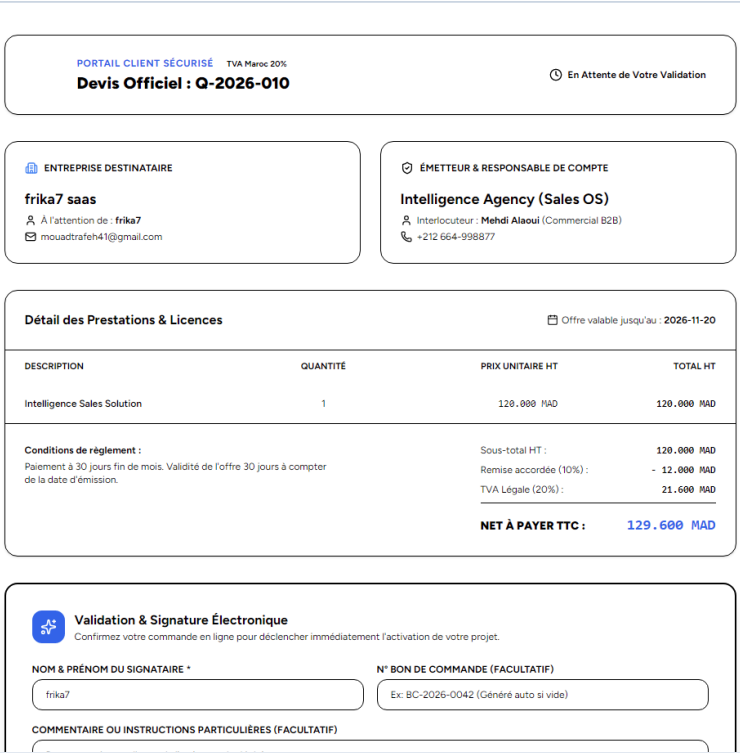


Figure 13: Public Quotation Portal — Itemized bill of materials, VAT calculation, and E-Signature form. `URL: /portal/quote/:token`

5. SECTION 2: Technical Specification for the 7 Sales AI Agents

For the Data & AI Developer: This section provides the exhaustive technical implementation blueprint for each of the 7 AI Agents. You must implement each agent following the exact trigger hooks, input payloads, prompt architectures, and output JSON schemas specified below.

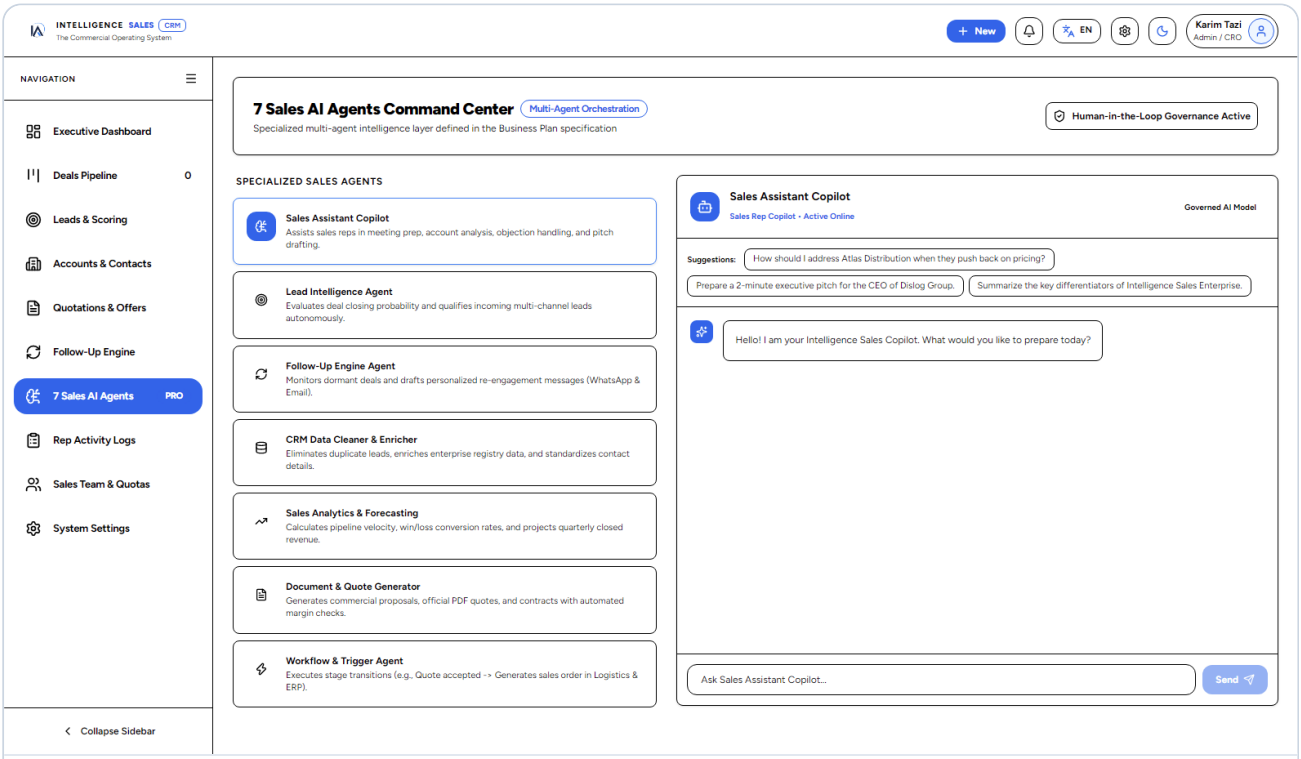


Figure 14: AI Agents Control Center — Central orchestrator managing the 7 autonomous commercial agents. `URL: /crm/ai-agents`

AI Agent #1: Lead Intelligence & Scoring Agent

CODE ID: LEAD_INTELLIGENCE

Mission: Instantly evaluates incoming commercial proposals, predicts conversion probability (0-100), tags temperature, and synthesizes an executive qualification rationale.

UI VISUAL ANCHOR
Figure 7: Leads Directory (/crm/leads)

TRIGGER HOOK
Event: POST /api/public/proposals & Lead::created

RECOMMENDED ML ENGINE
LightGBM Classifier + Claude 3.5 Sonnet / Gemini 1.5 Flash

TARGET LATENCY SLA
< 1,200 ms (Asynchronous Event Listener)

Input JSON Payload (From Laravel):

```
{
  "lead_id": 14,
  "company_name": "Atlas Agro SARL",
  "contact_name": "Sara Chraibi",
  "email": "sara@atlasagro.ma",
  "phone": "+212 661 99 88 77",
  "city": "Meknès",
  "estimated_volume": "200 000 - 500 000 MAD",
  "product_category": "Fresh Agri-Products Wholesale",
  "notes": "Urgent quote request for 500 tons seasonal supply."
}
```

Expected AI Output Schema (To be returned as pure JSON):

```
{
  "ai_score": 92,
  "temperature": "Hot",
  "score_reason": "High-volume wholesale demand with verified corporate domain (.ma). Immediate purchasing timeframe in a
  "risk_factors": ["Seasonal shelf-life sensitivity", "High initial credit exposure"],
  "recommended_rep_skills": ["Commodity Agri-Trade", "Fast Fulfillment Logistics"]
}
```

Database Mutation: Updates `leads.ai_score = 92` and `leads.score_reason = score_reason`.

5. AI Technical Blueprint (Cont.): Agents #2 & #3

AI Agent #2: Sales Assistant Copilot

CODE ID: SALES_ASSISTANT

Mission: 24/7 copilot for commercial sales reps. Generates real-time meeting preparation briefs, tactical objection handling talking points (especially against price pushback), and competitor counter-arguments.

UI VISUAL ANCHOR
Figure 4: Deal Detail Inspection Modal

TRIGGER HOOK
On-demand Rep click / Deal stage advancement

RECOMMENDED ML ENGINE
GPT-4o / Claude 3.5 Sonnet (RAG over Activity History)

CONTEXT WINDOW
Last 10 activities + Account financials + Deal items

Input JSON Payload:

```
{
  "deal_id": 8,
  "deal_title": "Agri-Tech Wholesale Supply",
  "stage": "negotiation",
  "amount_mad": 320000,
  "client_objection": "Client claims competitor offers 12% lower prices with 60-day credit terms.",
  "account_profile": {
    "name": "Maroc Agrum SA",
    "annual_revenue": "45M MAD",
    "credit_tier": "Tier A"
  }
}
```

Expected AI Output Schema:

```
{
  "ai_recommendation": "Do not slash prices directly. Propose split-delivery batching with 5% volume rebate tied to promp
  "talking_points": [
    "Emphasize guaranteed 24h cold-chain delivery reliability over cheaper uncertified rivals.",
    "Highlight full tax compliance with verifiable ICE & electronic invoicing."
  ],
  "counter_offer_terms": {
    "proposed_discount": 5,
    "payment_terms": "30 days end of month",
    "concession_condition": "Minimum order quantity 25 tons"
  }
}
```

Database Mutation: Updates `deals.ai_recommendation` and logs an activity note.

AI Agent #3: Smart Follow-Up Engine Agent

CODE ID: FOLLOW_UP

Mission: Enforces the 7-Day Inactivity Rule. Detects dormant opportunities, drafts personalized multi-channel re-engagement messages (WhatsApp & Email in EN/FR/AR), and requires human rep approval before sending.

UI VISUAL ANCHOR
Figure 9: Activities & Follow-Up Timeline

TRIGGER HOOK
Cron: Every night at 02:00 AM (`app:check-dormant-deals`)

DORMANCY THRESHOLD
Last interaction > 7 days

LANGUAGES
French (Français), Darija / Arabic (العربية), English

Expected AI Output Schema:

```
{
  "is_dormant": true,
  "days_inactive": 9,
  "whatsapp_draft_fr": "Bonjour Omar, suite à notre échange sur le devis DEV-2026-004, nos équipes ont pré-réservé votre
  "whatsapp_draft_ar": "خصوص عرض السعر الأخير، السلعة راه محجوزة في المستودع. واش ممكن نتواصلو هاد العشية لتأكيد الطلبية؟
  "email_subject": "Suivi de votre cotation DEV-2026-004 — Disponibilité des stocks",
  "urgency_rating": "High"
}
```

Database Mutation: Inserts draft into `activities` with `completed = false` waiting for rep approval.

5. AI Technical Blueprint (Cont.): Agents #4 & #5

AI Agent #4: CRM Data Cleaner & Enricher

CODE ID: CRM_DATA

Mission: Eliminates duplicate records, normalizes phone numbers to standard E.164 (`+212...`), validates Moroccan tax registries (ICE 15 digits, IF, RC, Patente), and flags data decay.

UI VISUAL ANCHOR
Figure 6 & 11: Accounts & Proposal Form

TRIGGER HOOK
On Account/Lead creation & Weekly cleanup job

TAX ID CHECKS
Moroccan ICE (15 digits), IF (8 digits), Patente

DEDUPLICATION METHOD
Fuzzy matching (Levenshtein + Token Sort) + ICE key

Expected AI Output Schema:

```
{
  "canonical_company_name": "MaroCraft Industries SARL AU",
  "normalized_phone": "+212 522 33 44 55",
  "is_duplicate": false,
  "duplicate_account_id": null,
  "ice_validation": {
    "ice_number": "001524398000045",
    "is_valid_format": true,
    "confidence_score": 0.99
  },
  "data_completeness_score": 95
}
```

AI Agent #5: Predictive Sales Analytics & Forecasting Agent

CODE ID: SALES_ANALYTICS

Mission: Projects quarterly closed revenue, calculates pipeline velocity, predicts deal slippage, and diagnoses win/loss root causes for executive leadership.

UI VISUAL ANCHOR
Figure 2: Executive Dashboard (/crm/dashboard)

TRIGGER HOOK
GET /api/analytics/kpis & Deal Stage Update

ALGORITHM
Survival Analysis & Probability-Weighted Pipeline Matrix

FORECAST HORIZON
Current Month, Quarter, and Trailing 12 Months

Expected AI Output Schema:

```
{
  "projected_quarterly_revenue_mad": 4850000,
  "forecast_confidence_interval": [4500000, 5200000],
  "pipeline_velocity_days": 21.4,
  "slippage_risk_deals": [
    {"deal_id": 5, "deal_title": "Atlas Logistics Supply", "risk": "High - Stage stagnant for 14 days"}
  ],
  "team_quota_attainment_forecast": 104.2
}
```

5. AI Technical Blueprint (Cont.): Agents #6 & #7

AI Agent #6: Document & Quote Generator Agent

CODE ID: DOCUMENT_AGENT

Mission: Automates itemized proposal generation, calculates gross margins, computes 20% legal Moroccan VAT (TVA), and enforces the 15% discount approval governance rule.

UI VISUAL ANCHOR
Figure 6: Quotations Page & Builder (/crm/quotes)

TRIGGER HOOK
POST /api/quotes (Quote Creation)

RULE ENGINE
Subtotal - Discount + 20% VAT = Total TTC

DISCOUNT THRESHOLD
> 15% requires Manager Approval (Locked Status)

Output Calculation Contract:

```
{
  "subtotal_ht": 225000.00,
  "discount_percent": 18,
  "discount_amount": 40500.00,
  "discount_approved": false,
  "requires_manager_approval": true,
  "approval_status": "Pending Approval",
  "net_ht": 184500.00,
  "tax_percent": 20,
  "tax_amount": 36900.00,
  "total_ttc": 221400.00,
  "currency": "MAD"
}
```

AI Agent #7: Autonomous Sales Workflow & Trigger Agent

CODE ID: AUTOMATION_AGENT

Mission: Event-driven sales workflow execution. When an external trigger occurs (e.g. client signs quote online, proposal submitted), automatically executes downstream state mutations: wins the deal, provisions accounts, dispatches logistics orders, and sends alerts.

UI VISUAL ANCHOR
Figure 13: Quote Signing Portal (/portal/quote/:token)

TRIGGER HOOK
Event: POST /api/public/quotes/:token/accept

IDEMPOTENCY KEY
Required (Prevents double stage progression)

DOWNSTREAM SYNC
Intelligence Logistique & ERP Sales Order Generation

Execution Flow Blueprint:

```
1. Client submits digital signature on /portal/quote/:token
2. automation_agent intercepts event:
   a) Sets quote.status = 'Accepted', logs signer_name & po_number
   b) Sets deal.stage = 'won', probability = 100%
   c) Creates Activity log entry for rep & manager
   d) Dispatches confirmation email to client with counter-signed PDF
   e) Emits webhook to ERP / Logistics API for order fulfillment
```

6. Developer Implementation Matrix & API Integration Contracts

The table below summarizes the technical specifications, communication protocols, and file locations for the Data/AI developer:

AGENT ID	AGENT NAME	TRIGGER EVENT	TARGET CONTROLLER	ASSOCIATED UI FILE
`lead_intelligence`	Lead Scoring	POST /api/public/proposals	`LeadController@publicStore`	`CrmLeads.jsx`
`sales_assistant`	Rep Copilot	On-Demand / Deal Inspection	`DealController@show`	`CrmPipeline.jsx`
`follow_up`	7-Day Follow-Up	Cron: app:check-dormant	`ActivityController@followUp`	`CrmActivities.jsx`
`crm_data`	Data Hygiene	POST /api/accounts & leads	`AccountController@store`	`CrmAccounts.jsx`
`sales_analytics`	Forecasting	GET /api/analytics/kpis	`AnalyticsController@kpis`	`CrmDashboard.jsx`
`document_agent`	Quote Engine	POST /api/quotes	`QuoteController@store`	`CrmQuotes.jsx`
`automation_agent`	Workflow Triggers	POST /public/quotes/:token/accept	`PublicQuoteController@accept`	`ClientQuotePortal.jsx`

Developer Confirmation Checklist

1. Confirm all 7 AI Agent code IDs match the backend database enumeration.
2. Ensure output JSON schemas follow exact key names (e.g. `ai_score`, `score_reason`, `ai_recommendation`).
3. Implement fallback heuristics for every AI in case the LLM API is unreachable (graceful degradation).
4. Enforce strict human-in-the-loop review for follow-up message dispatches and discount overrides.